

MP491 : Summer Exam 20/21

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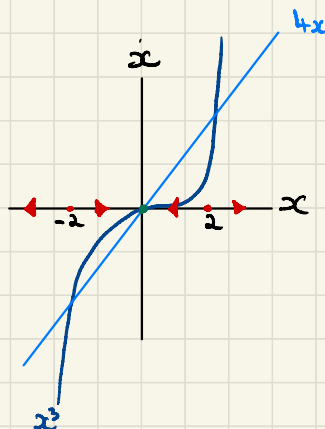


$$1(a) \quad \frac{dx}{dt} = \dot{x} = x(x^2 - 4)$$

Equilibrium points when $\dot{x} = 0$

$$x(x^2 - 4) = 0$$

$$x^* = 0, x^* = \pm 2$$



If $\dot{x} > 0$, moves right
 $x^3 - 4x > 0$
 $x^3 > 4x$

$x^* = 0$ is stable
 $x^* = -2, 2$ are unstable

If $\dot{x} < 0$, moves left
 $x^3 < 4x$

$$1(b) \quad \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Delta-Tau plane

$$\Delta = \left(\frac{3+i\sqrt{11}}{2} \right) \left(\frac{3-i\sqrt{11}}{2} \right) = \frac{20}{2} = 10$$

$$\tau = \frac{3+i\sqrt{11}}{2} + \frac{3-i\sqrt{11}}{2} = 3$$

$$\tau^2 - 4\Delta = -31$$

$$\tau^2 - 4\Delta < 0, \tau > 0, \Delta > 0$$

$\therefore$ Equilibrium point at (0,0) is an unstable spiral

Equilibrium point $(x, y) = (0, 0)$

$$\text{Coefficient matrix } A = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}$$

Eigenvalues: $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 3 \\ -1 & 1-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(1-\lambda) - (-1)(3) = 0$$

$$2 - 3\lambda + \lambda^2 + 3 = 0$$

$$\lambda^2 - 3\lambda + 5 = 0$$

$$\lambda_{1,2} = \frac{3 \pm \sqrt{9-20}}{2} = \frac{3 \pm i\sqrt{11}}{2}$$

Direction of Rotation

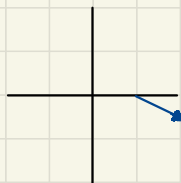
Let $x = 0 + \epsilon$

$$y = 0$$

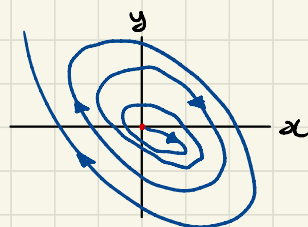
for $\epsilon > 0$

$$\dot{x} = 2\epsilon$$

$$y = -\epsilon$$



Phase-Plane portrait



$$(c) \quad \frac{dx}{dt} = X = 2x + 3y + 2x^2 - 6xy$$

$$\frac{dy}{dt} = Y = x - 2y + 3y^2 - 4xy$$

$$(i) \quad \text{Hamiltonian is } \frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} = 0$$

$$\frac{\partial X}{\partial x} = 2 + 4x - 6y \quad \frac{\partial Y}{\partial y} = -2 + 6y - 4x$$

$$\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} = 2 + 4x - 6y - 2 + 6y - 4x = 0 \quad \therefore \text{the system is Hamiltonian}$$

$$(ii) \quad \text{Hamiltonian Function } H(x, y): \frac{\partial H}{\partial x} = -\frac{dy}{dt}, \frac{\partial H}{\partial y} = \frac{dx}{dt}$$

$$\frac{\partial H}{\partial x} = -Y(x, y) = -x + 2y - 3y^2 + 4xy$$

$$\frac{\partial H}{\partial y} = X(x, y) = 2x + 3y + 2x^2 - 6xy$$

$$H(x, y) = \frac{x^2}{2} + 2yx - 3y^2x + 2x^2y + f(y)$$

$$H(x, y) = 2xy + \frac{3}{2}y^2 + 2x^2y - 3xy^2 + f(x)$$

$$H(x, y) = \frac{x^2}{2} + 2yx - 3y^2x + 2x^2y + 2xy + \frac{3}{2}y^2 + 2x^2y - 3xy^2 + C$$

$$\text{Set } C=0: H(x, y) = 4x^2y - \frac{x^2}{2} - 6xy^2 + 4xy + \frac{3}{2}y^2$$

$$(d) \quad F = x_{n+1} = x_n(x_n - 2)$$

Fixed points occur when $x_{n+1} = x_n = \bar{x}$

$$\bar{x} = \bar{x}(\bar{x} - 2)$$

$$\bar{x} = \bar{x}^2 - 2\bar{x}$$

$$\bar{x}^2 - 3\bar{x} = 0$$

$$\bar{x}(\bar{x} - 3) = 0$$

$$\bar{x} = 0, \bar{x} = 3$$

Linear stability:

$$F_x = 2x - 2$$

$$F_x(0) = 2(0) - 2 = -2 < 1 \dots 0 \text{ is a stable fixed point}$$

$$F_x(3) = 2(3) - 2 = 4 > 1 \dots 3 \text{ is an unstable fixed point}$$

2. (a)

$$\frac{dx}{dt} = x^3 - (a-3)x$$

(i) Equilibrium points when $\dot{x} = 0$

$$x^3 - (a-3)x = 0$$

$$x(x^2 - (a-3)) = 0$$

$$x_1^* = 0, \quad x_2^* = \pm\sqrt{a-3}$$

$$x_2^* \text{ exists}$$

if $a > 3$

$$x_2^* \text{ exists}$$

if $a = 3$

$$x_2^* = x_1^*$$

if $a < 3$

x_2^* doesn't exist

when $\dot{x} > 0$, moves right

$$x^3 - (a-3)x > 0$$

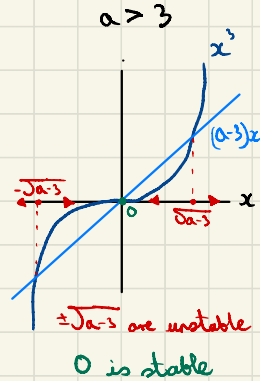
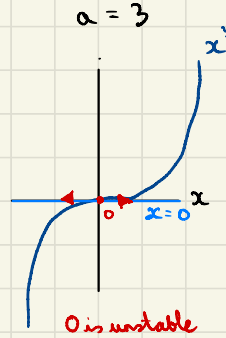
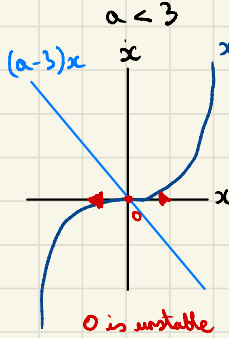
$$x^3 > (a-3)x$$

when $\dot{x} < 0$, moves left

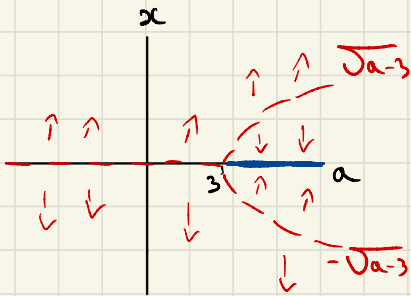
$$x^3 < (a-3)x$$

Phase portrait

(ii)



(iii) Bifurcation Diagram



(b)

$$\frac{dx}{dt} = x + (3-a)y$$

$$\frac{dy}{dt} = x + y$$

Coefficient Matrix

$$A = \begin{bmatrix} 1 & 3-a \\ 1 & 1 \end{bmatrix}$$

Delta-Tau plane

$$\tau = \text{tr}(A) = 2$$

$$\Delta = \det(A) = 1 - (3-a) = -2+a$$

$$\tau^2 - 4\Delta = 4 - 4(-2+a)$$

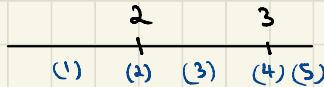
$$= 4 + 8 - 4a$$

$$= 12 - 4a$$

Critical points

$$\Delta = 0 \text{ when } a = 2$$

$$\tau^2 - 4\Delta = 0 \text{ when } a = 3$$



The nature of the equilibrium point (0,0) as a varies:

(1) $a < 2$: $\tau > 0$, $\Delta < 0$, $\tau^2 - 4\Delta > 0$: saddle point(2) $a = 2$: $\tau > 0$, $\Delta = 0$, $\tau^2 - 4\Delta > 0$: whole line of fixed points(3) $2 < a < 3$: $\tau > 0$, $\Delta > 0$, $\tau^2 - 4\Delta > 0$: unstable node(4) $a = 3$: $\tau > 0$, $\Delta > 0$, $\tau^2 - 4\Delta = 0$: unstable degenerate node(5) $a > 3$: $\tau > 0$, $\Delta > 0$, $\tau^2 - 4\Delta < 0$: unstable spiral
 $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ has $\lambda = 1$, $\lambda > 0$ and only one real eigenvector

3

$$\frac{dx}{dt} = \dot{x} = y - x^2 + 1$$

$$\frac{dy}{dt} = \dot{y} = x^2 + y - 1$$

(a) Equilibrium Points: $\frac{dx}{dt} = 0, \frac{dy}{dt} = 0$

$$\text{Jacobian: } J = \begin{pmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} \end{pmatrix}$$

$$y - x^2 + 1 = 0$$

$$x^2 + y - 1 = 0$$

$$y = x^2 - 1$$

$$x^2 + (x^2 - 1) - 1 = 0$$

$$2x^2 - 2 = 0$$

$$x^2 - 1 = 0$$

$$(x+1)(x-1) = 0$$

$$x = -1, x = 1$$

$$J = \begin{pmatrix} -2x & 1 \\ 2x & 1 \end{pmatrix}$$

$$J(-1,0) = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$\tau = 3, \Delta = 4, \tau^2 - 4\Delta = -7$$

$(-1,0)$ is an unstable spiral

$$J(1,0) = \begin{pmatrix} -2 & 1 \\ 2 & 1 \end{pmatrix}$$

$$\tau = -1, \Delta = -4, \tau^2 - 4\Delta = 17$$

$(1,0)$ is a saddle point

$$e_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

(b) Isoclines: $K = \frac{dy}{dx} = \frac{x^2 + y - 1}{y - x^2 + 1}$

$K = 0$ horizontal, $\frac{dy}{dt} = 0$

$$x^2 + y - 1 = 0$$

$$y = 1 - x^2$$

→ when $\frac{dx}{dt} > 0$

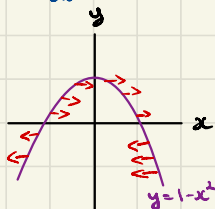
$$y - x^2 + 1 > 0$$

$$(1 - x^2) - x^2 + 1 > 0$$

$$-2x^2 + 2 > 0$$

$$x^2 - 1 < 0$$

$$-1 < x < 1$$



$K = \infty$ vertical, $\frac{dx}{dt} = 0$

$$y - x^2 + 1 = 0$$

$$y = x^2 - 1$$

→ when $\frac{dy}{dt} > 0$

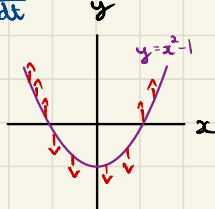
$$x^2 + y - 1 > 0$$

$$x^2 + (x^2 - 1) - 1 > 0$$

$$2x^2 - 2 > 0$$

$$(x+1)(x-1) > 0$$

$$x < -1, x > 1$$



$K = 1, 45^\circ, \frac{dx}{dt} = \frac{dy}{dt}$

$$y - x^2 + 1 = x^2 + y - 1$$

$$2x^2 - 2 = 0$$

$$(x+1)(x-1) = 0$$

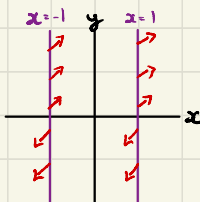
$$x = -1, x = 1$$

→ when $\frac{dy}{dt} > 0$

$$x^2 + y - 1 > 0$$

$$1 + y - 1 > 0$$

$$y > 0$$



$K = -1, -45^\circ, \frac{dy}{dt} = -\frac{dx}{dt}$

$$x^2 + y - 1 = -(y - x^2 + 1)$$

$$2y = 0$$

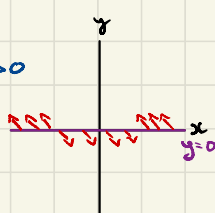
$$y = 0$$

→ when $\frac{dy}{dt} > 0$

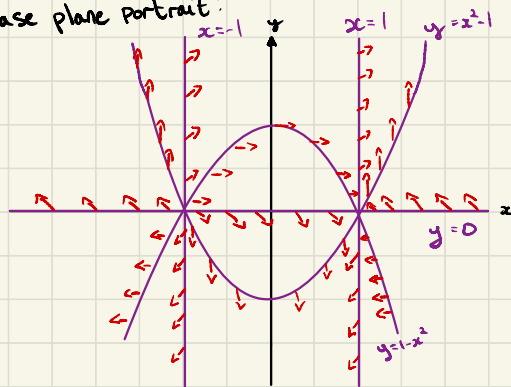
$$x^2 + y - 1 > 0$$

$$x^2 - 1 > 0$$

$$x < -1, x > 1$$



(c) Phase plane portrait:



We can see an unstable spiral at $(-1,0)$, and saddle point at $(1,0)$